

Discrete Structures

Lecture # 14

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TREE

A **tree** is a **connected graph** that does not contain any **nontrivial circuit**. (it is **circuit-free**).

A **trivial circuit** is one that consists of a **single vertex**.

EXAMPLE



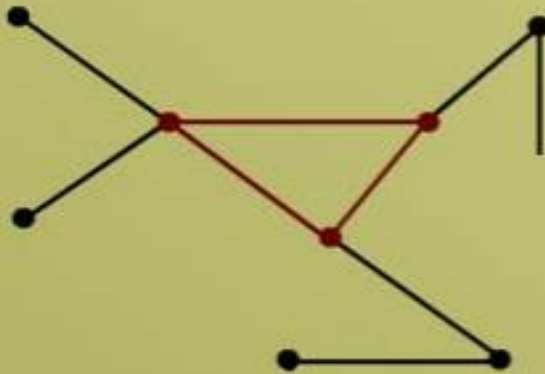
TREE

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TREE

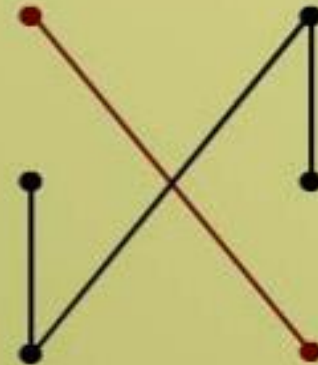


TREE

EXAMPLES OF NON TREES



(a) Graph with a circuit.



(b) Disconnected graph.



(c) Graph with a circuit.

SOME SPECIAL TREES

1. TRIVIAL TREE

A **graph** that consists of a **single vertex** is called a **trivial tree** or **degenerate tree**.

2. EMPTY TREE

A **tree** that does not have any **vertices or edges** is called an **empty tree**.

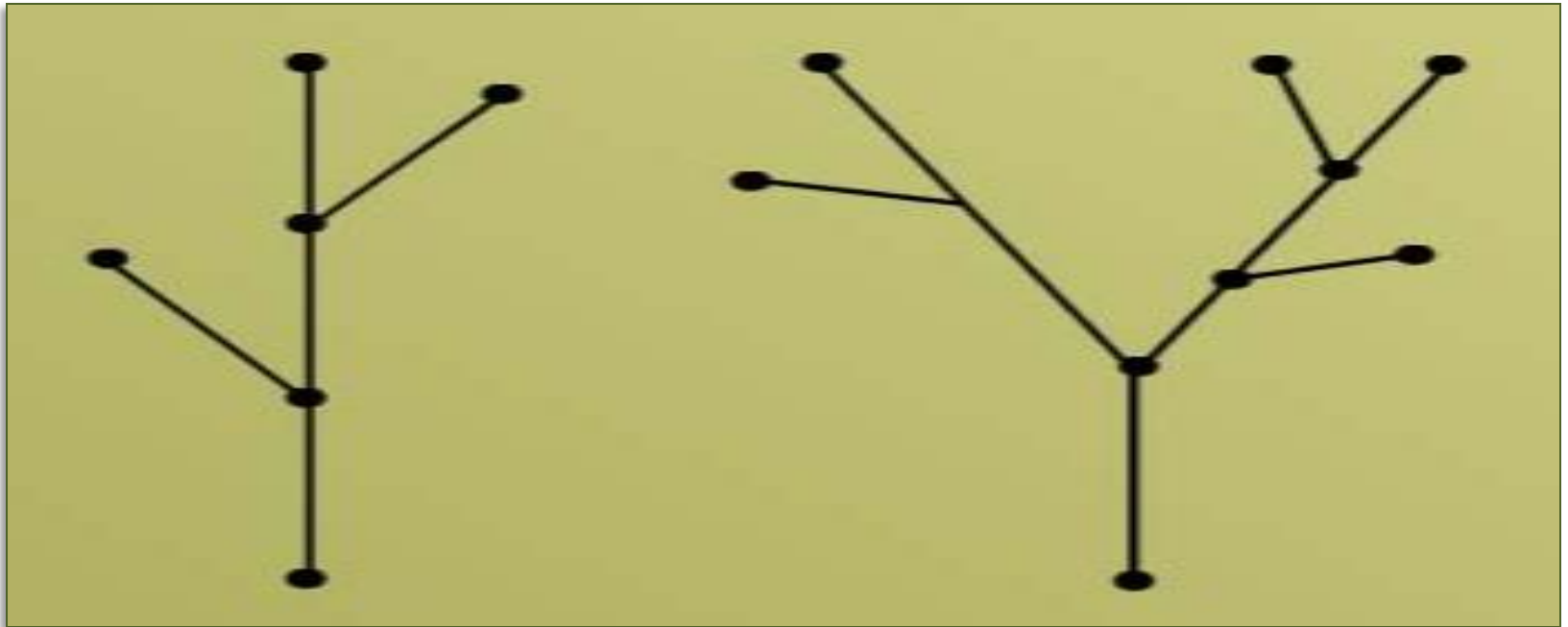
SOME SPECIAL TREES

FOREST:

A **graph** is called a **forest** if, and only if, it is **circuit-free**.

Hence, the **connected components** of a **forest** are **trees**.

FOREST



Forest has only **two trees** in it.

PROPERTIES OF TREES

1. A **tree** with **n vertices** has **$n - 1$ edges**.
2. Any **connected graph** with **n vertices** and **$n - 1$ edges** is a **tree**.
3. A **tree** has no **nontrivial circuit**; but if **one new edge** is added to it, then the resulting **graph** has exactly **one nontrivial circuit**.

PROPERTIES OF TREES

4. A **tree** is connected, but if any **edge** is deleted from it, then the **resulting graph** is **not connected**.
5. Any **tree** that has more than one **vertex** has at least **two vertices** of **degree 1**.
6. A **graph** is a **tree** iff there is a **unique path** between any two of its **vertices**.

EXERCISE

Explain why **graphs (Tree)** with the given specification do not **exist**.

1. Tree with **twelve vertices** and **fifteen edges**.
2. Trees with **five vertices** and total **degree 10**.

SOLUTION:

1. Any tree with 12 vertices will have $12 - 1 = 11$ edges.

Hence solution of first is not possible.

SOLUTION

Tree with **five vertices** and total degree **10**.

Any **tree** with **five vertices** will have $5 - 1 = 4$ **edges**.

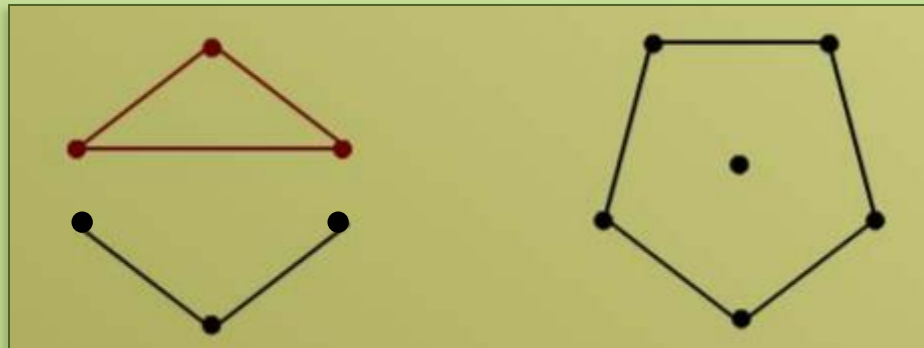
We are given **total degree** of graph is **10**. So it must have **edges** $10/2 = 5$.

The two conditions contradict each other.

SOLUTION

Draw a graph with **six vertices**, **five edges** that is not a **tree**.

SOLUTION:

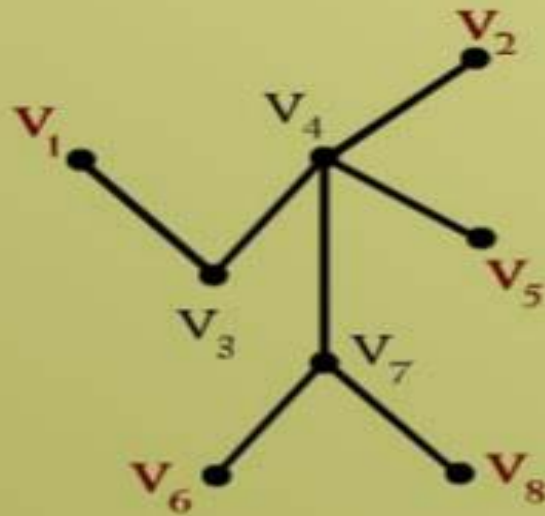


First is not **tree** because it is **not connected** and also has a **circuit** similarly for second.

TERMINAL AND INTERNAL VERTEX

A vertex of degree '1' in a tree is called terminal vertex or a leaf (leaf has no children) and a vertex of degree greater than '1' in a tree is called an internal vertex or a branch vertex.

EXAMPLE



v_1, v_2, v_5, v_6 and v_8 are **terminal vertices**.

v_3, v_4, v_7 are **internal vertices**.

ROOTED TREE

A **rooted tree** is a **tree** in which **one vertex** is **distinguished** from the others and is called the **root**.

The **level of a vertex** is the number of **edges** along the **unique path** between it and the **root**.

The **height of a rooted tree** is the maximum level to **any vertex** of the **tree**.

ROOTED TREE

The **children** of any **internal vertex v** are all those **vertices** that are **adjacent to v** and are **one level farther** away from the **root** than **v** . If **w** is a **child** of **v** , then **v** is called the **parent** of **w** .

Two **vertices** that are **both children** of the **same parent** are called **siblings**.

Given **vertices v** and **w** , if **v** lies on the unique path between **w** and the **root**, then **v** is an **ancestor** of **w** and **w** is a **descendant** of **v** .

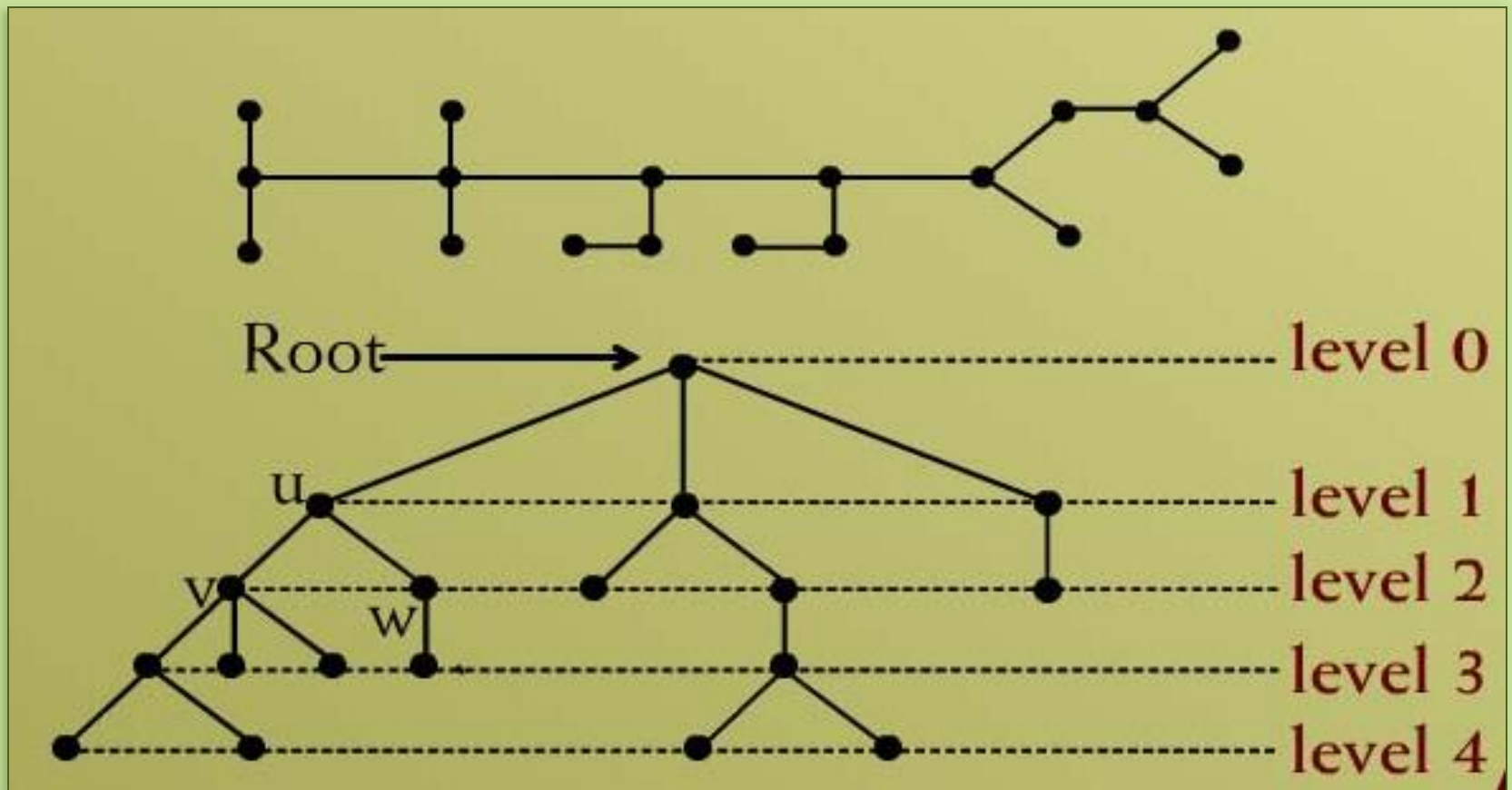
ROOTED TREE

The root is an internal vertex unless it is the only vertex in the graph, and in that case it will be a leaf.

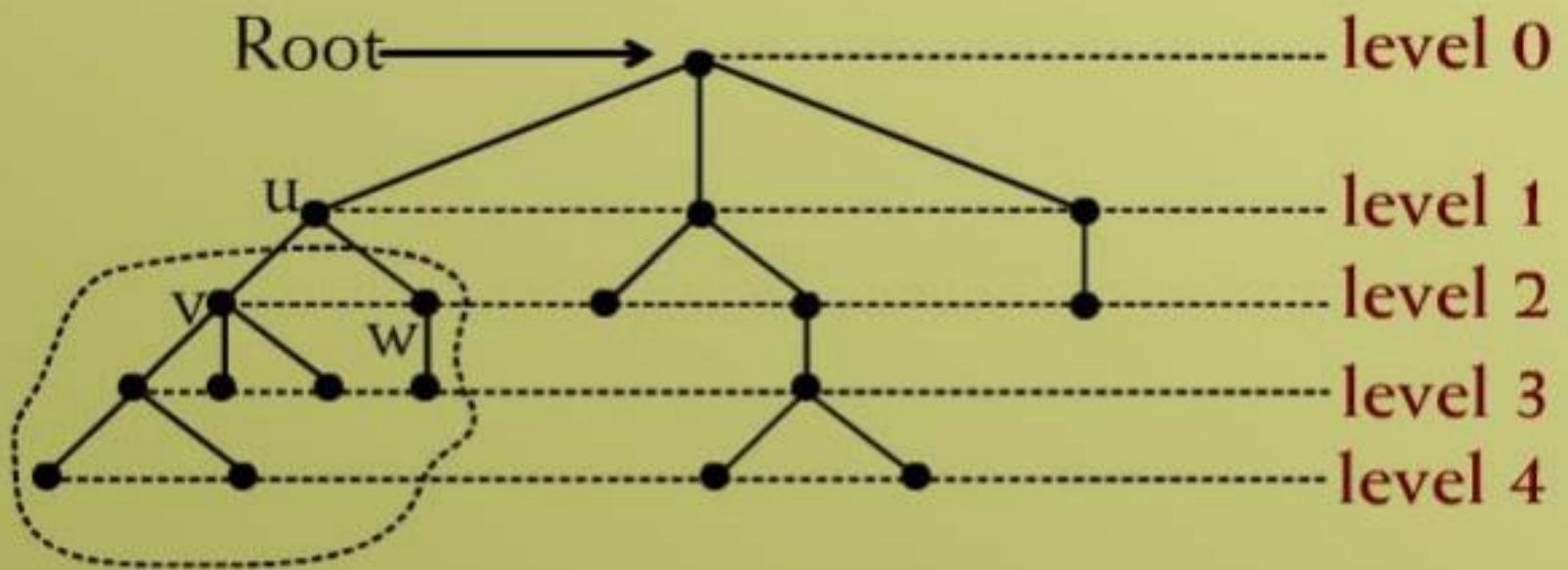
Subtree

If a is a vertex in a tree, the subtree with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

EXAMPLE

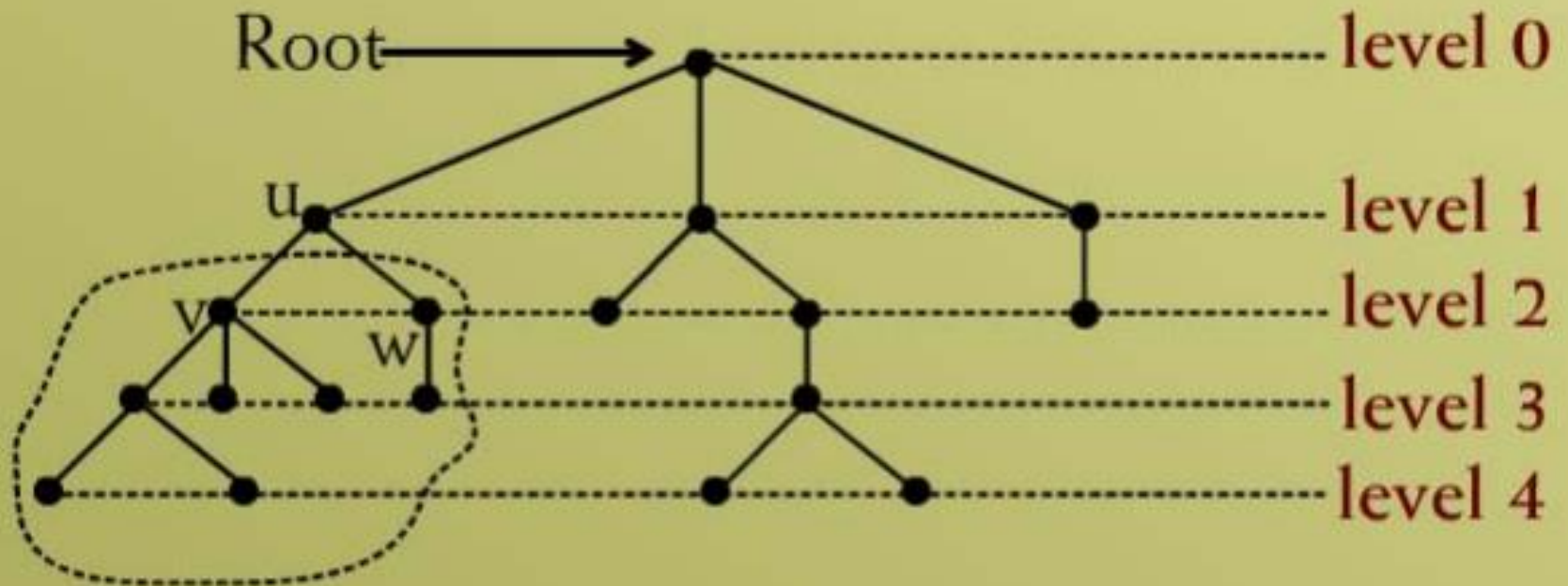


EXAMPLE



Vertices in enclosed region are descendants of u , which is an ancestor of each.

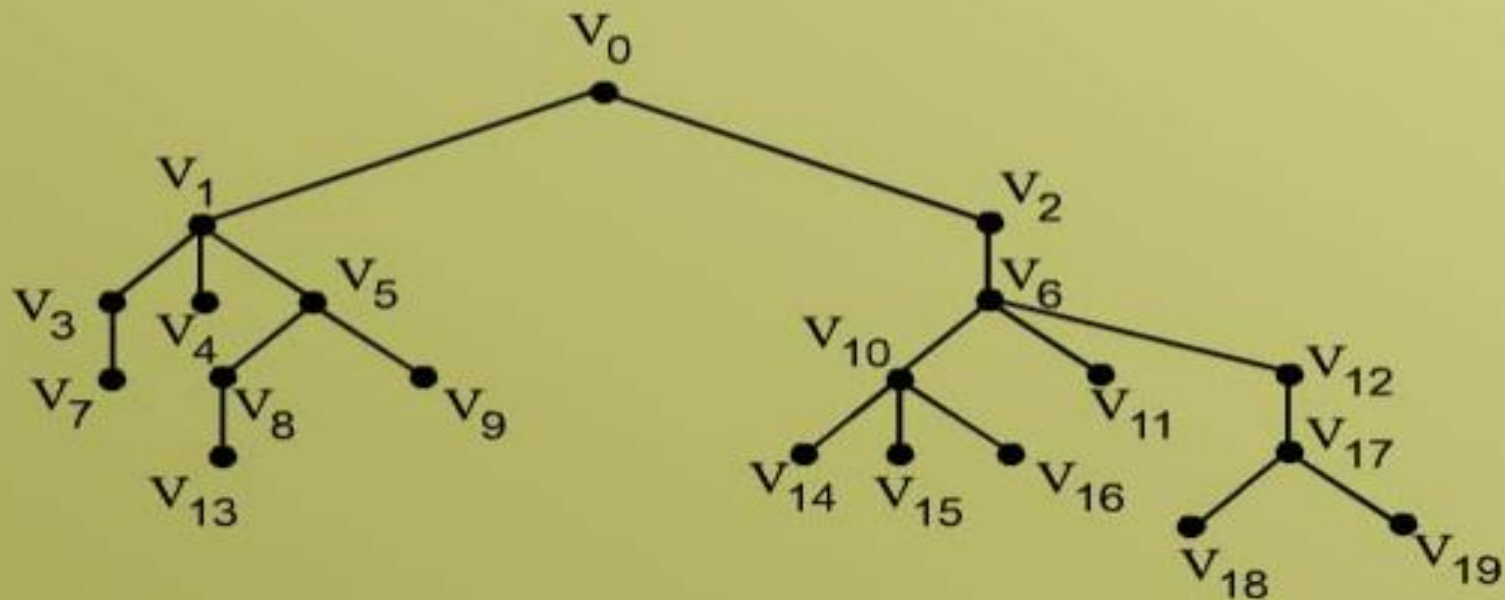
EXAMPLE



v is a **child** of u , u is the **parent** of v , v and w are **sublings**, $\text{height} = 4$.

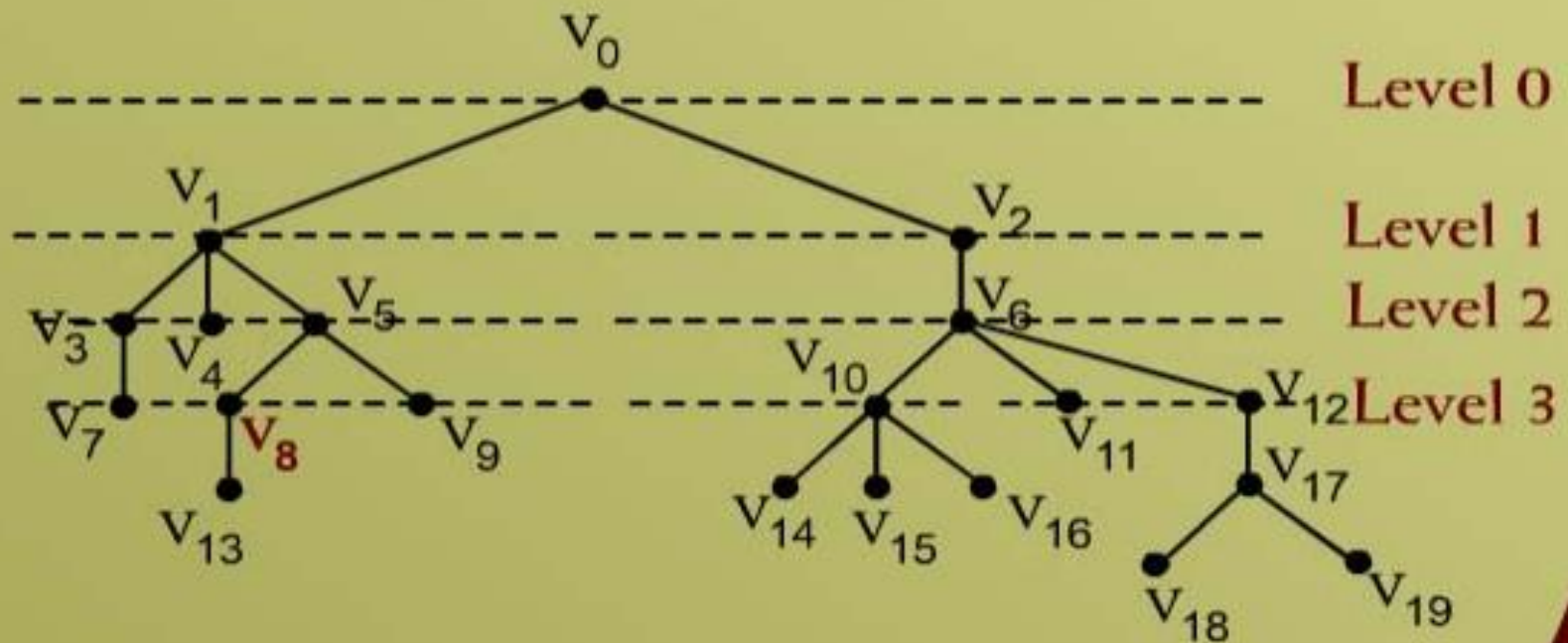
EXERCISE

Consider the **rooted tree** shown below with root v_0 .



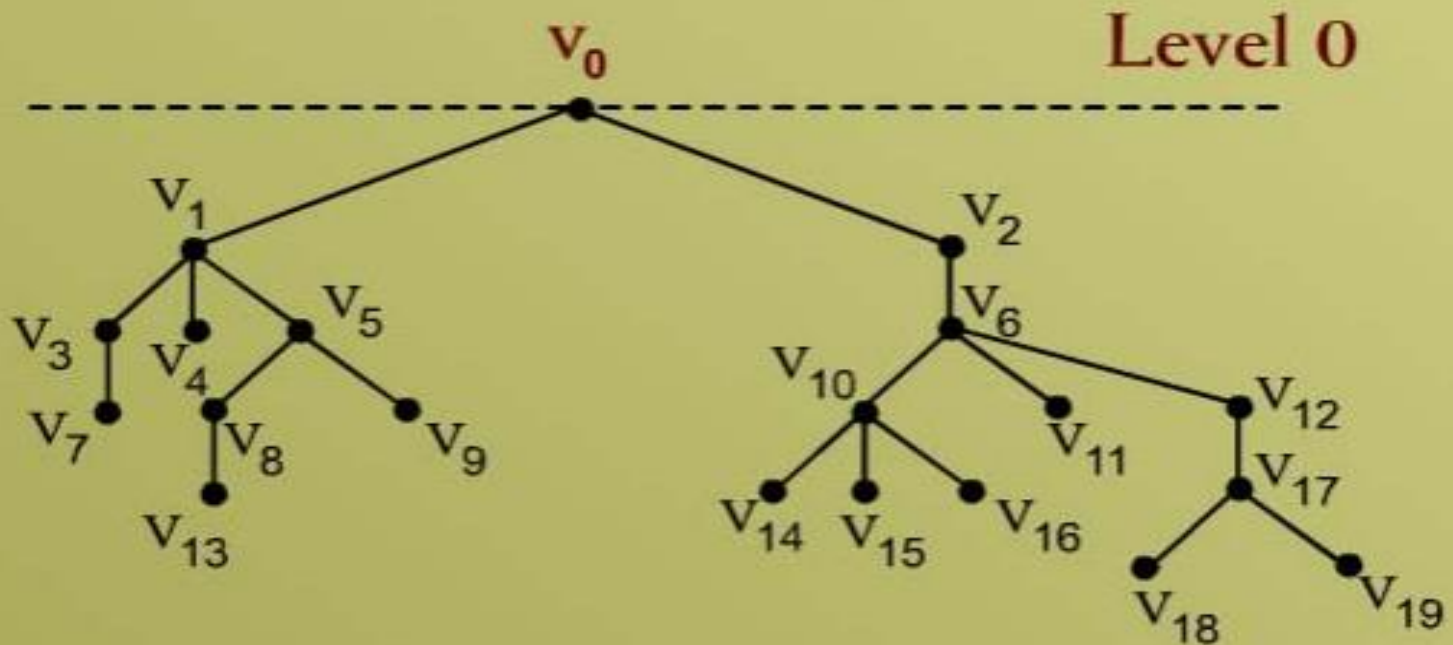
EXERCISE

a. What is the **level** of v_8 ?



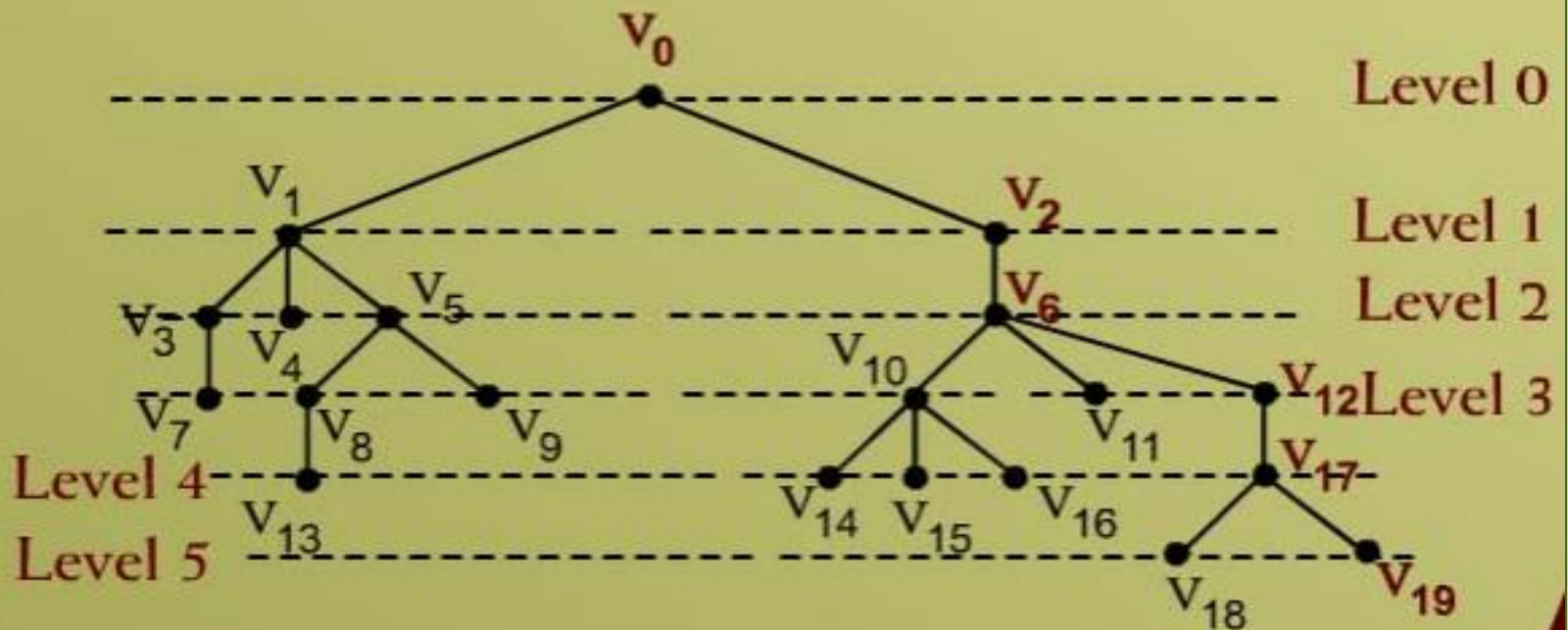
EXERCISE

b. What is the **level** of v_0 ?



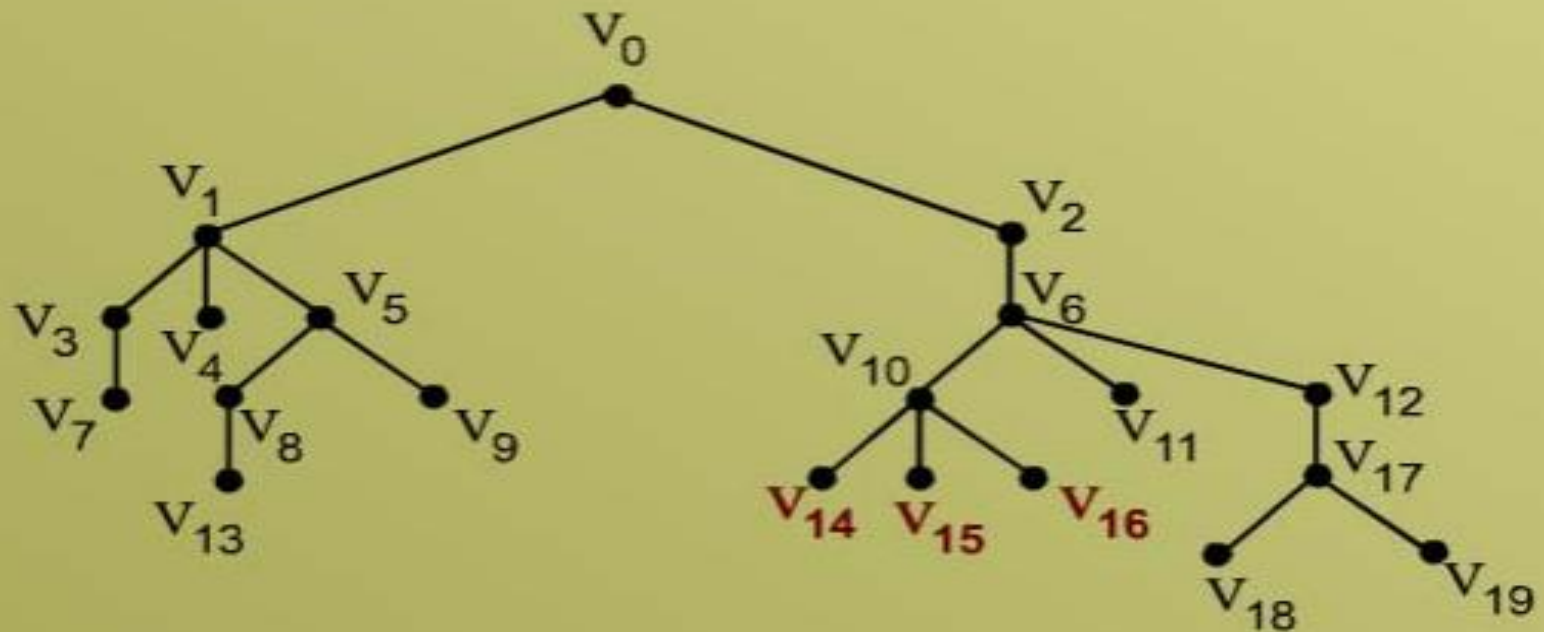
EXERCISE

c. What is the **height** of this tree?



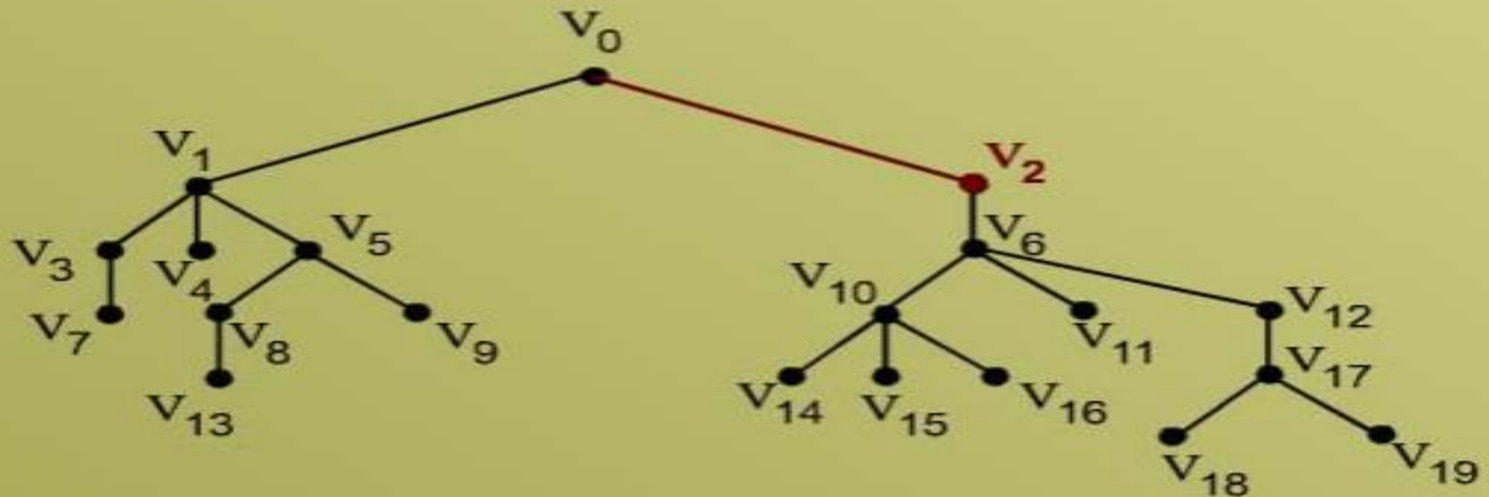
EXERCISE

d. What are the **children** of v_{10} ?



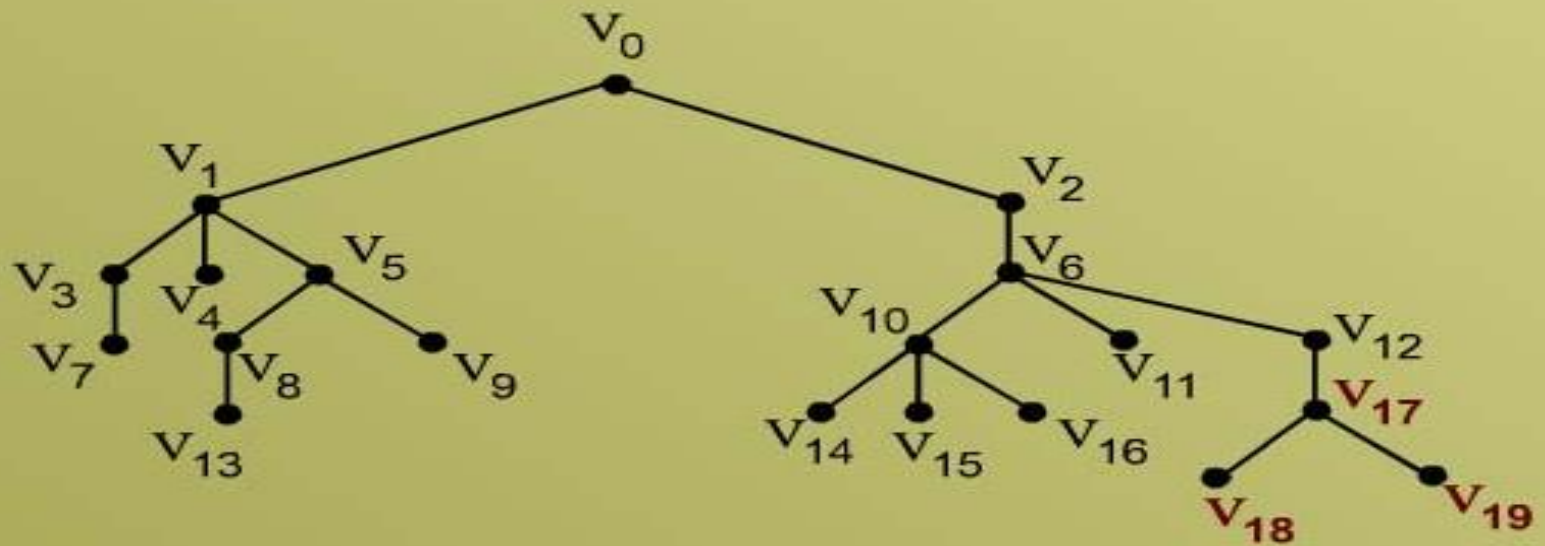
EXERCISE

e. What are the **siblings** of v_1 ?



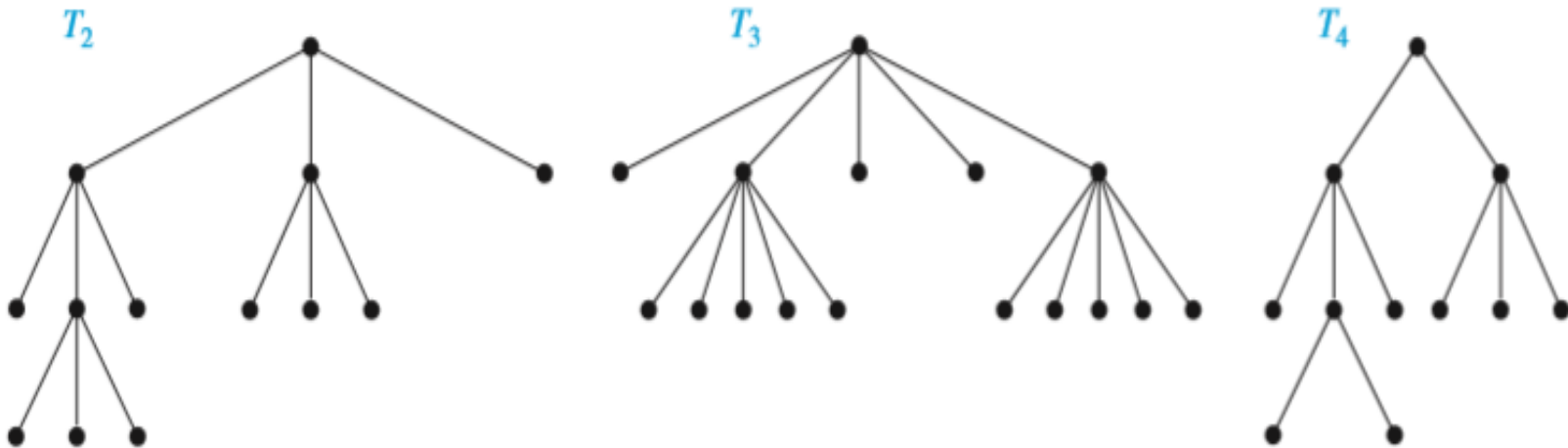
EXERCISE

f. What are the **descendants** of v_{12} ?



m-ary TREE

- A rooted tree is called *m*-ary tree if every internal vertex has no more than *m* children.
- The tree is called *full m*-ary tree if every internal vertex has exactly *m* children.
- An *m*-ary tree with $m=2$ is called a *binary tree*.



BINARY TREE

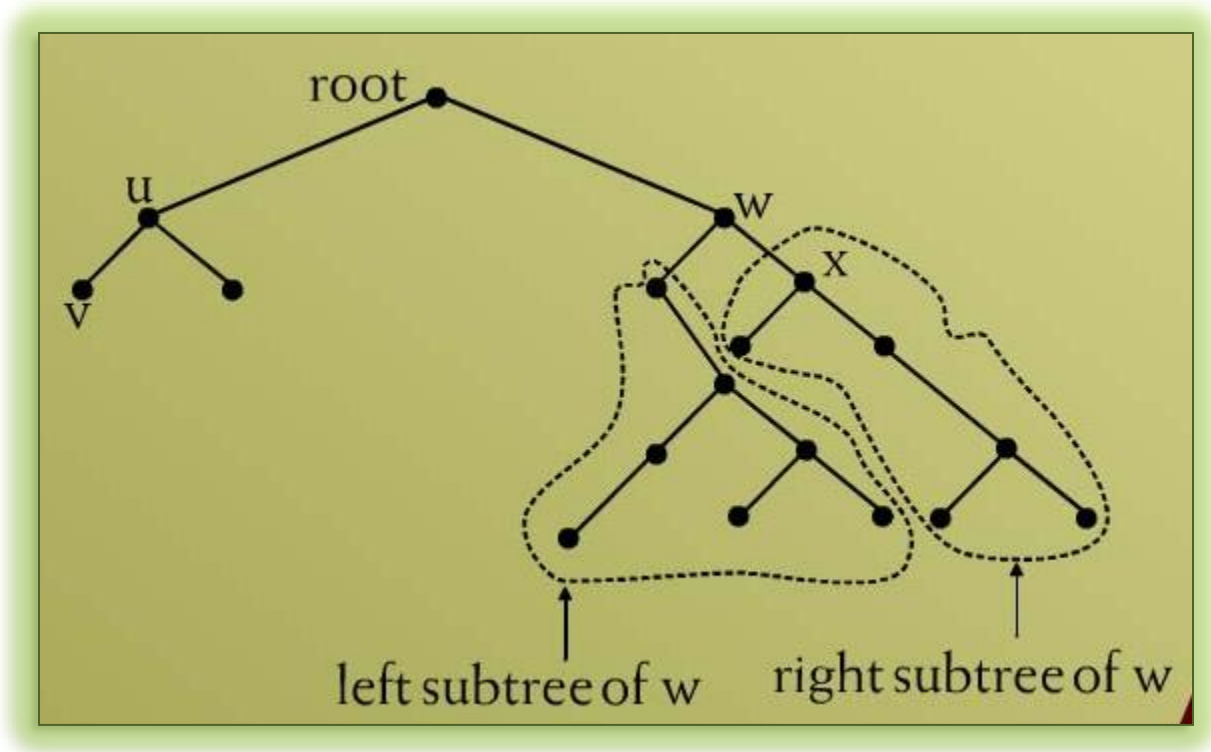
A **binary tree** is a **rooted tree** in which every **internal vertex** has at **most two children**.

Every child in a **binary tree** is designated either a **left child** or a **right child**.

FULL BINARY TREE

A full binary tree is a binary tree in which each internal vertex has exactly two children.

EXAMPLE



THEOREM

A full m -ary tree with k internal vertices contains $n = mk + 1$ vertices.

1. If k is a positive integer and T is a full binary tree with k internal vertices, then T has a total of $2k + 1$ vertices and has $k + 1$ terminal vertices.

THEOREM

A full m -ary tree with

1. n vertices has $k = (n-1)/m$ internal vertices and $l = [(m-1)n + 1]/m$ leaves.
2. k internal vertices has $n = mk + 1$ vertices and $l = (m-1)k + 1$ leaves.
3. l leaves has $n = (ml-1)/(m-1)$ vertices and $k = (l-1)/(m-1)$ internal vertices.

THEOREM

There are at most m^h leaves (terminal vertices) in an m -ary tree of height h .

THEOREM

2. If T is a **binary tree** that has **t terminal vertices** and **height h** , then

$$t \leq 2^h$$

Equivalently,

$$\log_2 t \leq h$$

EXERCISE

Explain why **graphs** with the given **specification** do not exist.

1. **Full binary tree with nine vertices and five internal vertices.**
2. **Binary tree with height 4 and eighteen terminal vertices.**

SOLUTION

1. Any **full binary tree** with **five internal vertices** has **six terminal vertices**, for a total of **eleven**, not **nine vertices** in all.

Thus there is **no full binary tree** with the given properties.

2. Any **binary tree** of **height 4** has at most **$2^4 = 16$** terminal vertices.

Hence, there is **no binary tree** that has **height 4** and **eighteen terminal vertices**.

EXERCISE

Draw a **full binary tree** with **seven vertices**.

SOLUTION:

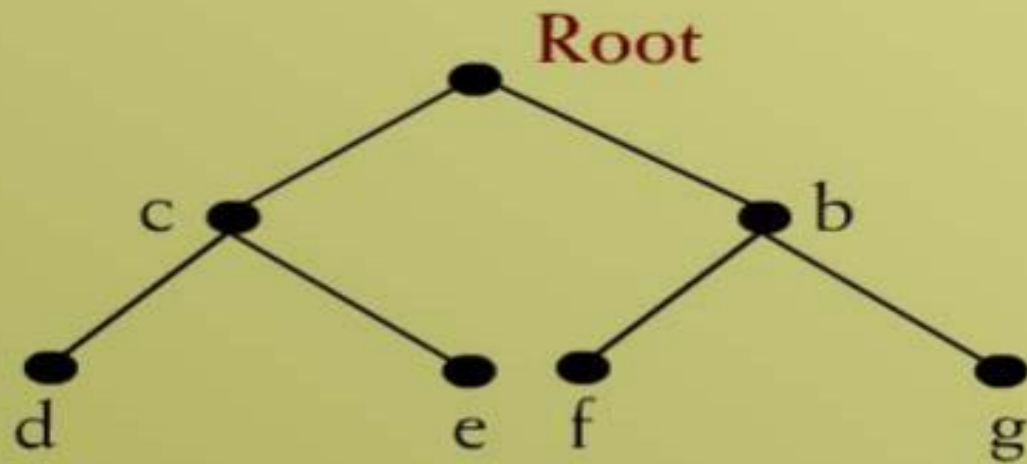
$$\text{Total vertices} = 2k + 1 = 7$$

$$\Rightarrow k = 3$$

Hence, number of **internal vertices** = $k = 3$

Number of **terminal vertices** = $k + 1 = 4$

EXERCISE

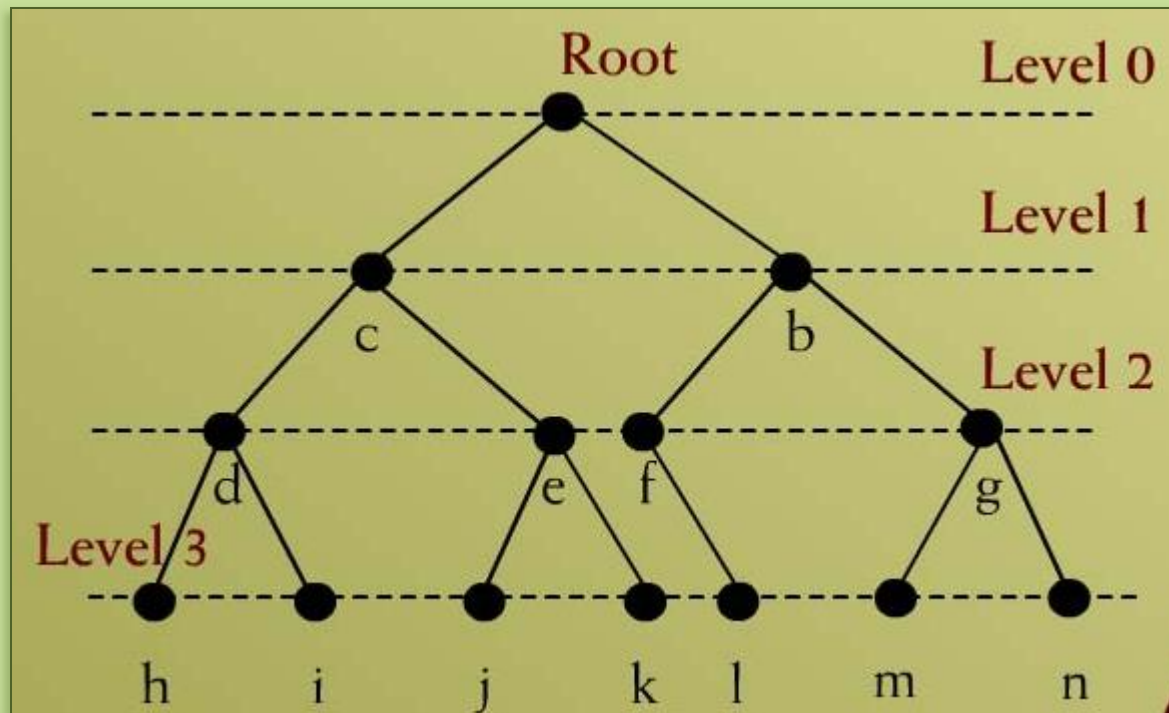


Which is required **full binary tree** with **seven vertices**.

EXAMPLE

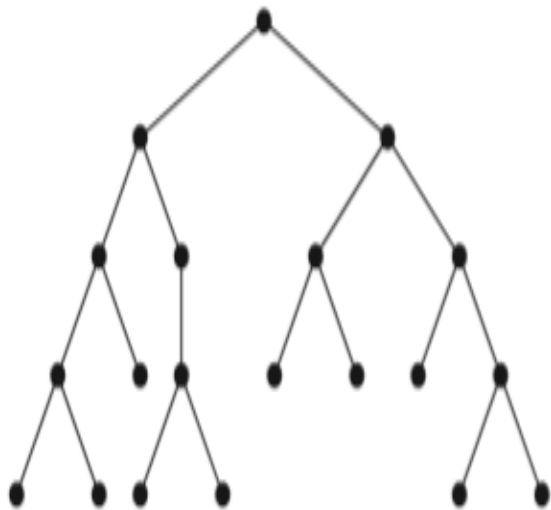
Draw a **binary tree** with **height 4 (level 3)**
and having **seven terminal vertices**.

SOLUTION



Balanced Rooted Tree

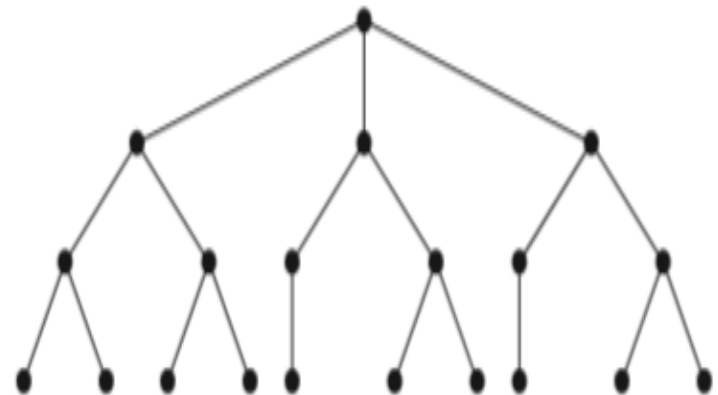
A rooted m -ary tree of height h is balanced if all leaves are at levels h or $h-1$.



T_1



T_2



T_3